

# Systematical errors and Barlow test

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In one of the previous exercises, it was discussed how to determine the statistical error of the measurement, e.g. by using the subsampling technique. Statistical errors are unavoidable and typically result from the finite amount of data available for the analysis. However, other types of errors can occur during the measurement, which can originate from the malfunctioning of the detector, corrupted data, etc. These errors are called systematical errors, and, unlike statistical errors, they cannot be suppressed by increasing the dataset size. Since statistical and systematical errors are fundamentally different, they are reported separately in an experiment.

In an actual experiment, it's not always easy to disentangle statistical and systematical errors. Especially in small datasets, variations due to large statistical fluctuations can be wrongly attributed to a certain degree to systematical errors. This problem is resolved with the so-called *Barlow test*, which is the subject of this exercise. The text in this exercise is reproduced and adapted with permission from M. Sc. thesis of Marcel Lesch in Ref. [1], while a more detailed exposure can be found in Refs. [2, 3].

The typical analysis in high-energy physics commences with the definition of default event and track selection criteria (i.e. default *cuts* for short), which leads to the default measurement and its statistical uncertainty,  $y_{\text{default}} \pm \sigma_{\text{default}}$ , of some observable of interest. To estimate systematical error, one of the default values in the event or track selection criteria is changed and the analysis is redone with this variation. As such, the results related to this single change of a parameter are referred to as the *trial* (denoted by  $y_{\text{trial}}$ ). To study whether the divergence between the trial and the initial default results  $y_{\text{default}}$  is of significance, their difference is calculated as a first step. It is defined as

$$Diff = y_{\text{trial}} - y_{\text{default}} . \quad (1)$$

To determine its statistical significance, the error of the trial  $\sigma_{\text{trial}}$  in comparison to the one of the default analysis  $\sigma_{\text{default}}$  has to be accounted for. The importance of this lies in the statistical influence of the used trials. In general, when a selection cut is tightened, less data is used for the analysis leading to an increased statistical error for the trial, while a looser cut leads to the opposite effect in general. To account for this, the so-called *error on the difference* is used:

$$Err = \sqrt{|\sigma_{\text{trial}}^2 - \sigma_{\text{default}}^2|} . \quad (2)$$

In the last equation, the minus sign has to be used because the datasets used for the trial and default are correlated to each other as they only differ in one selection criterion while being part of the larger, default dataset. Would the two datasets be uncorrelated to each other, a plus sign would have to be used in Eq. (2). Given the difference and the error in difference, the so-called  $\sigma_{\text{Barlow}}$  can be computed as

$$\sigma_{\text{Barlow}} = \frac{Diff}{Err} . \quad (3)$$

The  $\sigma_{\text{Barlow}}$  quantifies the statistical significance of the performed trial. If it is found to be greater than a certain value, the trial is classified as a statistically relevant deviation. In this exercise, a critical value of  $\sigma_{\text{Barlow}} = 2$  is used i.e. every trial with  $\sigma_{\text{Barlow}} > 2$  is counted as statistically important.

As a last step, the systematical error has to be computed. For this, the so-called *relative variation* is used and systematical error is computed as

$$\sigma_{\text{sys}} = \frac{Diff}{y_{\text{default}}} . \quad (4)$$

In case more independent trials have been performed, the final systematical error of all trials is then obtained as the square root of all the individual systematical errors added in quadrature, e.g. for  $n$  trials:

$$\sigma_{\text{sys,final}} = \sqrt{\sigma_{\text{sys},1}^2 + \sigma_{\text{sys},2}^2 + \dots + \sigma_{\text{sys},n}^2} . \quad (5)$$

This general formalism is now applied for concrete example in this exercise. Consider the univariate function  $f(x)$  defined by the following formula:

$$f(x) = Nxe^{-x}, \quad x \in [0, 1), \quad (6)$$

where  $N$  is normalization constant.

**Question 1:** Determine analytically  $N$  from the normalization  $\int_0^1 f(x)dx = 1$ . After that, calculate analytically the true mean value  $\mu_x$  (also denoted as expectation value  $E[x]$  or  $\langle x \rangle$ ) from the formula  $\mu_x \equiv \int_0^1 xf(x)dx$ .

**Question 2:** Implement the code in ROOT which will treat the function  $f(x)$  defined in Eq. (6) as a probability density function (p.d.f.) on the interval  $0 < x < 1$ , where  $x$  is variable (see examples in <http://root.cern.ch/root/html/TF1.html>). It's perfectly fine if you re-use and adapt in this part some code developed for previous exercises.

**Question 3:** Sample  $x$  from the p.d.f.  $f(x)$  and inject randomly among sampled values the “outliers”, i.e. take randomly values for  $x$  also from interval  $[1, 2)$  which is outside of the initial phase space of  $x$  defined in Eq. (6). Simulate outliers with some suitable frequency, e.g. after 10 healthy events, inject in the dataset one outlier event, or something similar. In this context, these “outliers” are nonphysical events, and they have to be removed from the results either by introducing special selection criteria, i.e. special cuts, or if that is not feasible, they have to be assigned to systematical error.

**Question 4:** From the sampled values of  $x$ , calculate  $\langle x \rangle$  and its statistical uncertainty  $\sigma$ , taking into account all values of  $x$  (including the outliers). This is the default measurement  $\langle x \rangle_{\text{default}}$ . Then, repeat the procedure, but take into the average  $\langle x \rangle$  only those values of  $x$  which satisfy the condition  $0 \leq x < 1$  (in this context, this is your analysis “cut”). This is the trial measurement  $\langle x \rangle_{\text{trial}}$ .

**Question 5:** Repeat Step 4 for several choices of total number of samplings  $N_{\text{events}}$ . For each value of  $N_{\text{events}}$ , calculate  $\sigma_{\text{Barlow}}$  from Eq. (3), and make a plot  $\sigma_{\text{Barlow}}(N_{\text{events}})$ . How many events you have to simulate (or in this exercise, how many times you have to sample  $x$ ), to reach the condition  $\sigma_{\text{Barlow}} > 2$ ? After you reached that condition, how you report your final result? Hint: Use the format  $\langle x \rangle \pm \sigma_{\text{stat}} \pm \sigma_{\text{sys}}$ .

## References

- [1] M. Lesch, “Analyses of Symmetry Plane Correlations in Pb–Pb Collisions at 2.76 TeV with ALICE at the LHC,” M. Sc. thesis (2021), Technical University of Munich.
- [2] R. Barlow, “Systematic errors: Facts and fictions,” [arXiv:hep-ex/0207026 [hep-ex]].
- [3] R. Barlow, “Statistics: A Guide to the Use of Statistical Methods in the Physical Sciences,” Manchester Physics Series, (1989), WileyBlackwell.